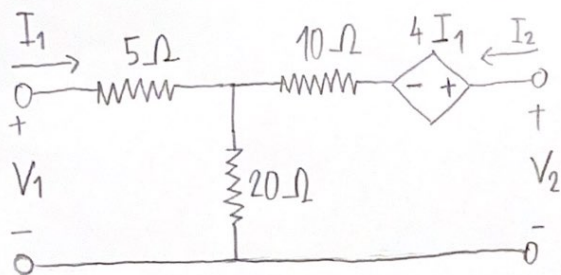


EE 201 Assignment #5

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$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

When $I_1 = 0$

* Dependent source has 0 voltage difference between its terminals

$$V_2 = 10 I_2 + 20 I_2 = 30 I_2 \Rightarrow Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = 30 \Omega$$

$$V_1 = 20 I_2 \Rightarrow Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} = 20 \Omega$$

When $I_2 = 0$

$$V_2 = 20 I_1 + 4 I_1 = 24 I_1 \Rightarrow Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} = 24 \Omega$$

$$V_1 = 5 I_1 + 25 I_1 \Rightarrow Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = 25 \Omega$$

$$Z_{11} = 25 \Omega, Z_{12} = 20 \Omega, Z_{21} = 24 \Omega, Z_{22} = 30 \Omega$$